

DBTHON 26 – Database Systems Hackathon

Objective: To develop an innovative database system for solving a real-world problem in a domain of your choice.

Your solution must:

- a) Identify a specific real-world problem involving data management;
- b) Design an appropriate database schema/model;
- c) Implement database operations using suitable DBMS technologies;
- d) Incorporate at least one meaningful database innovation;
- e) Demonstrate appropriate SQL queries, constraints, transactions, indexing, security mechanisms, or other relevant DBMS concepts;
- f) Evaluate the proposed solution using suitable performance, accuracy, scalability, security, or data-quality metrics.

The solution should demonstrate a clear relationship between the problem, database design, innovation, implementation, and measurable outcome.

What each team must demonstrate?

Component	Demonstration Requirement
1. Problem Definition	Clearly explain the problem, target users, and motivation
2. Domain	Identify and justify the selected application domain
3. Existing System	Explain the existing approach and its limitations
4. Database Design	Demonstrate the ER diagram and relational schema
5. Innovation	Demonstrate the specific technical innovation introduced
6. Novelty	Clearly explain what is new or different from existing/conventional approaches
7. Prototype	Demonstrate a working DBMS-based prototype
8. Evaluation	Demonstrate results/evidence showing the improvement achieved

Innovation about your solution

Describe your proposed solution by explaining the existing approach used to address the problem, its key limitations, the innovation you are proposing to overcome those limitations, how the innovation will be technically implemented using database concepts or technologies, and the measurable improvement expected from the proposed solution.

Each team must incorporate at least one meaningful database-related innovation into its proposed solution.

Possible areas include:

- Intelligent query processing
- Adaptive indexing
- Natural Language to SQL
- Vector databases
- Graph databases
- Temporal databases
- Distributed databases
- Privacy-preserving databases
- Intelligent database security
- Automated database tuning
- Real-time/stream processing
- Edge databases
- Database–AI integration
- Database–IoT integration
- Energy-aware database processing
- Data-quality and anomaly detection
- Other database innovations proposed by the team

These are examples, not restrictions. Teams are encouraged to propose their own ideas.

Novelty Requirement

The project must contain a feature, mechanism, architecture, algorithm, or integration that is meaningfully different from a conventional database implementation. Simply using a modern technology does not automatically constitute novelty.

Students should clearly establish:

1. Existing Approach
2. Limitation
3. Proposed Approach
4. Novel Component
5. Measurable Benefit

Example

Domain: Agriculture

Problem: Large volumes of IoT sensor data make continuous monitoring inefficient.

Existing Limitation: Conventional systems periodically query stored sensor records, resulting in delayed detection.

Innovation: Real-time database-level anomaly detection.

Novelty: Integrating continuous stream processing with database-level anomaly detection so that abnormal sensor patterns can be identified without repeatedly scanning the complete historical dataset.

Validation: Compare response latency, query processing time, and unnecessary data processing against a conventional approach

Novelty will be evaluated based on:

- Distinctiveness of the proposed approach
- Relevance to the identified problem
- Technical depth
- Integration with DBMS concepts
- Evidence supporting the claimed improvement
- Quality of implementation and demonstration

Students are encouraged to develop original solutions rather than simply combining existing technologies without a clear technical purpose.

Evaluation Rubric (30 Marks)

CO1: Design the structure and operation of the relational data model.

CO2: Examine the data requirements of the real world and design a database management system.

S. No.	Evaluation Component	Marks	COs
1	Problem Identification & Domain Relevance	4	CO2
2	Database Design & Modeling	5	CO1, CO2
3	DBMS Implementation & Technical Depth	5	CO1
4	Innovation	4	CO1, CO2
5	Novelty & Differentiation	5	CO2
6	SDG Alignment & Societal Impact	2	CO2
7	Validation & Measurable Improvement	3	CO1, CO2

8	Technology Readiness Level (TRL) & Demonstration	2	CO1, CO2
Total		30	